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Tetrasubstituted allenes via the palladium-catalysed kinetic resolution of propargylic alcohols using a supporting ligand

Wei-Feng Zheng[1], Wanli Zhang[1], Chaofan Huang[1], Penglin Wu[1], Hui Qian[1*], Lei Wang[2], Yin-Long Guo[2*] and Shengming Ma[1,2*]

Efficient enantioselective construction of fully substituted allenes from simple starting materials is still a fundamental challenge due to the difficulty to discriminate between the four substituents at the 1- and 3-positions of the three-carbon axis of chirality. Here, we report a straightforward catalytic asymmetric synthesis of tetrasubstituted 2,3-allenoic acids from readily available racemic propargylic alcohols. Enabled by the co-catalysis of palladium and a Brønsted acid in the presence of a commercially available chiral ligand (DTBM-SEGphos) and an achiral monophosphine supporting ligand (PPh_3), the kinetic resolution of propargylic alcohols proceeded efficiently in the presence of water and 1 atm CO, affording tetrasubstituted 2,3-allenoic acids in excellent enantioselectivity and atom economy with a good functional group compatibility. Performing a second kinetic resolution on the unreacted alcohol gave access to the other enantiomer of the product. These allenes are precursors to compounds with quaternary carbon centres and other chiral tetrasubstituted allene building blocks, which are of great interest.

It is well recognized that quaternary carbon centres exist widely in natural products, bioactive compounds and drugs (Fig. 1a). When the four substituents are different, a quaternary carbon stereogenic centre is generated. Because of the resultant increased diversity and conformational constraint, the creation of such a centre is of great relevance to the fields of bioscience and medicinal chemistry. However, the asymmetric construction of such quaternary carbon centres remains a challenge and of high current scientific and practical importance[1,2]. Allenes, meanwhile, have unique properties when compared to alkenes and alkynes due to their intrinsic three-carbon axial chirality. They have been found in natural and non-natural bioactive molecules[3] as well as functional materials[4], and are considered to be versatile building blocks, ligands and catalysts for organic synthesis[5–15].

Recently, much effort has been devoted to the generation of this unique linear axial chirality[16–25]. Optically active tetrasubstituted allenes are of fundamental interest, providing an alternative approach to creating the challenging and very important chiral quaternary carbon centres (up to three) (Fig. 1b)[1,2], but it is still challenging to access these allenes[26–30]. Although asymmetric addition to (di)cyclohexenynones, as developed by Hayashi[31], Sun[32] and their colleagues (Fig. 1c), and the enantioselective trapping of some rather specific allenyl anions with different electrophiles, as developed by Maruoka[33], Miller[34], Feng/Liu[35,36] and List[37], have been reported (Fig. 1d), racemic propargylic alcohols, a class of very common chemicals, would, in principle, be the most straightforward starting point to creating optically active tetrasubstituted allenes (Fig. 1e). However, no such protocol has been reported, hinting that traditional catalysts may not work for this purpose.

Here, we report our developments in this area—efficient metal-catalysed successive kinetic resolutions for the synthesis of both enantiomers of tetrasubstituted 2,3-allenoic acids[33,37] from readily available racemic tertiary propargylic alcohols by applying a commercially available chiral ligand, DTBM-SEGphos, with Brønsted acid, forming a transient leaving group (Fig. 1f). The key here is the addition of PPh_3 as the supporting ligand. These synthetically versatile allenoic acid products may be easily converted to other useful functionalized allenes and cyclic compounds with a quaternary carbon stereogenic centre[5,7,10,12–14].

Results

Optimization of reaction conditions. We initiated the study by the traditional strategy of screening various commercially available chiral ligands for the asymmetric carboxylation reaction of 2-phenyloct-3-yn-2-ol **1a** with $(PhO)_2POOH$ to form a transient leaving group (for details see Supplementary Fig. 1 and equation (1)). After screening nearly all different types of commercially available ligands with different reaction parameters, as expected we identified that (*R*)-DTBM-SEGphos was the best ligand, affording (*R*)-**2a** in 38% yield with merely 33% e.e. and a 42% yield of achiral 2,4-dienoic acid (*E*)-**2a′** (Table 1, entry 3). We then found that the addition of supporting ligand, PPh_3, led to a dramatic and unexpected enhancement in enantioselectivity (42% yield, 80% e.e., Table 1, entry 4), and only 3% yield of by-product (*E*)-**2a′**. As a comparison, when 6 mol% of supporting ligand PPh_3 was added with (*R*)-SEGphos, the enantioselectivity was also improved to 16% from 3% (Table 1, entries 1 and 2).

We next conducted an extensive optimization of the reaction conditions to improve the e.e. for the formation of allenoic acid (*R*)-**2a** (for details see Supplementary Tables 1–5). The enantioselectivity was improved to 85% by increasing the loading of water to 20 equiv. and PPh_3 to 20 mol% (Supplementary Table 1, entry 1). $(PhO)_2POOH$ was found to be most efficient in activating the hydroxyl group after examining different Brønsted acids

[1]Research Centre for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai, China. [2]State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai, China. *e-mail: qian_hui@fudan.edu.cn; ylguo@sioc.ac.cn; masm@sioc.ac.cn

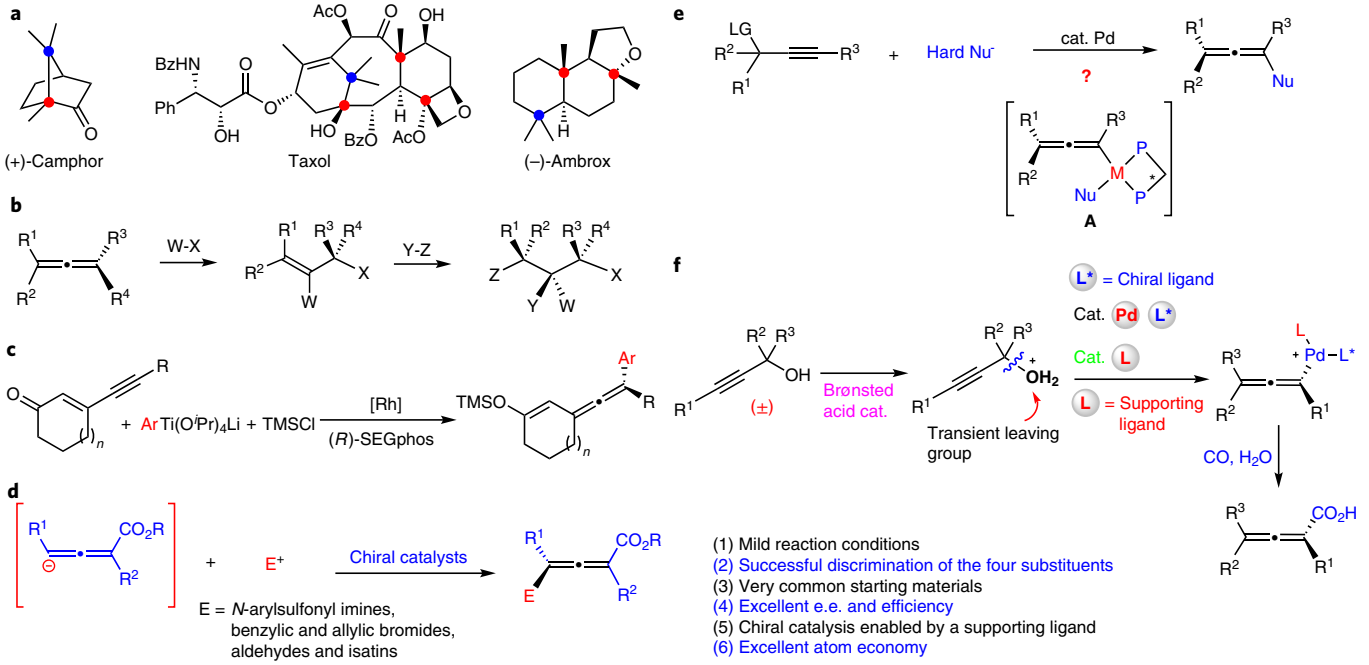
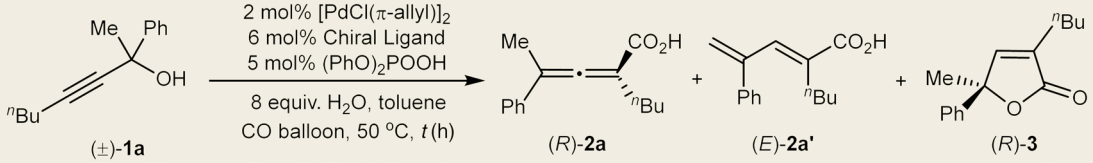


Fig. 1 | Approaches to tetrasubstituted allenes. a, Important compounds with quaternary carbon centre(s). Blue circles are achiral quaternary carbon centres, red circles are chiral quaternary carbon centres. **b**, The allene approach. **c**, Approaches to tetrasubstituted allenes via conjugate addition to (di) enynones[31,32]. TMS, trimethylsilyl. **d**, Approaches to tetrasubstituted allenes via allenylation of electrophiles[33–37]. **e**, Envisioned catalytic allenylation of tertiary propargylic alcohols or derivatives. **f**, This work: carboxylation of tertiary propargylic alcohols.

Table 1 | The effect of supporting ligand PPh_3



Entry	Chiral ligand	Supporting ligand	t (h)	(*R*)-2a		(*E*)-2a′ yield[a] (%)	(*R*)-3		Recovery of 1a[a] (%)
				Yield[a] (%)	e.e.[b] (%)		Yield[a] (%)	e.e.[b] (%)	
1	(*R*)-SEGphos	None	18	15	3	1	8	-	72
2	(*R*)-SEGphos	6 mol% PPh_3	1	19	16	1	-	-	79
3	(*R*)-DTBM-SEGphos	None	18	38	−33	42	5	47	3
4	(*R*)-DTBM-SEGphos	6 mol% PPh_3	1	42	−80	3	-	-	53

Reaction conditions: **1a** (0.5 mmol), [Pd(*π*-allyl)Cl]$_2$ (2 mol%), chiral ligand (6 mol%), $(PhO)_2POOH$ (5 mol%) and H_2O (8 equiv.) in toluene (1 ml) at 50 °C with a CO balloon unless otherwise noted. [a]Determined by ^{1}H NMR analysis using dibromomethane as the internal standard. [b]Determined by HPLC analysis.

(Supplementary Table 1, entries 1–4; for the results with more types of Brønsted acid, see Supplementary Table 4). The solvent effect was also investigated (Supplementary Table 1, entries 5–8). The reaction in chlorobenzene performed similarly to that in toluene, offering similar yield and enantioselectivity. 1,2-dichloroethane (DCE), tetrahydrofuran (THF), and acetonitrile (MeCN) are not suitable for this transformation. Very brief examinations of other reaction parameters, such as temperature and the loading of the Brønsted acid and palladium catalyst, led to the following optimal reaction conditions: 2 mol% of $PdCl_2$, 4.8 mol% of (*S*)-DTBM-SEGphos, 20 mol% of $(PhO)_2POOH$, 20 mol% of PPh_3 and 20 equiv. of H_2O in toluene at −5 °C under 1 atm of CO (Supplementary Table 1, entry 16), affording the desired allenoic acid (*R*)-**2a** in 41% NMR yield with 94% e.e.

Substrate scope. With the optimized conditions, we next examined the scope of this straightforward approach for chiral 2,3-allenoic acid synthesis. As shown in Fig. 2, a variety of racemic tertiary propargylic alcohols bearing electron-neutral, electron-rich and electron-deficient aryl moieties all smoothly underwent this transformation, forming tetrasubstituted chiral 2,3-allenoic acids ((*S*)-**2a** to (*S*)-**2j**) in good yields with over 90% e.e. Steric hindrance slightly affects the enantioselectivity because the e.e. for the substrates bearing the methyl group at the *o*-, *m*- and *p*-positions decreased in order from (*S*)-**2b** to (*S*)-**2d**. It is worth noting that when R^2 is an alkyl group, such as ^{t}Bu, Cy and ^{n}Pr, the reaction also afforded the desired 2,3-allenoic acid (*S*)-**2k**, (*S*)-**2l**, (*S*)-**2m** and (*S*)-**2n** with high enantioselectivities. The R^1 group may comprise carbon chains ranging from C_3 to C_8 and various functional groups, such as

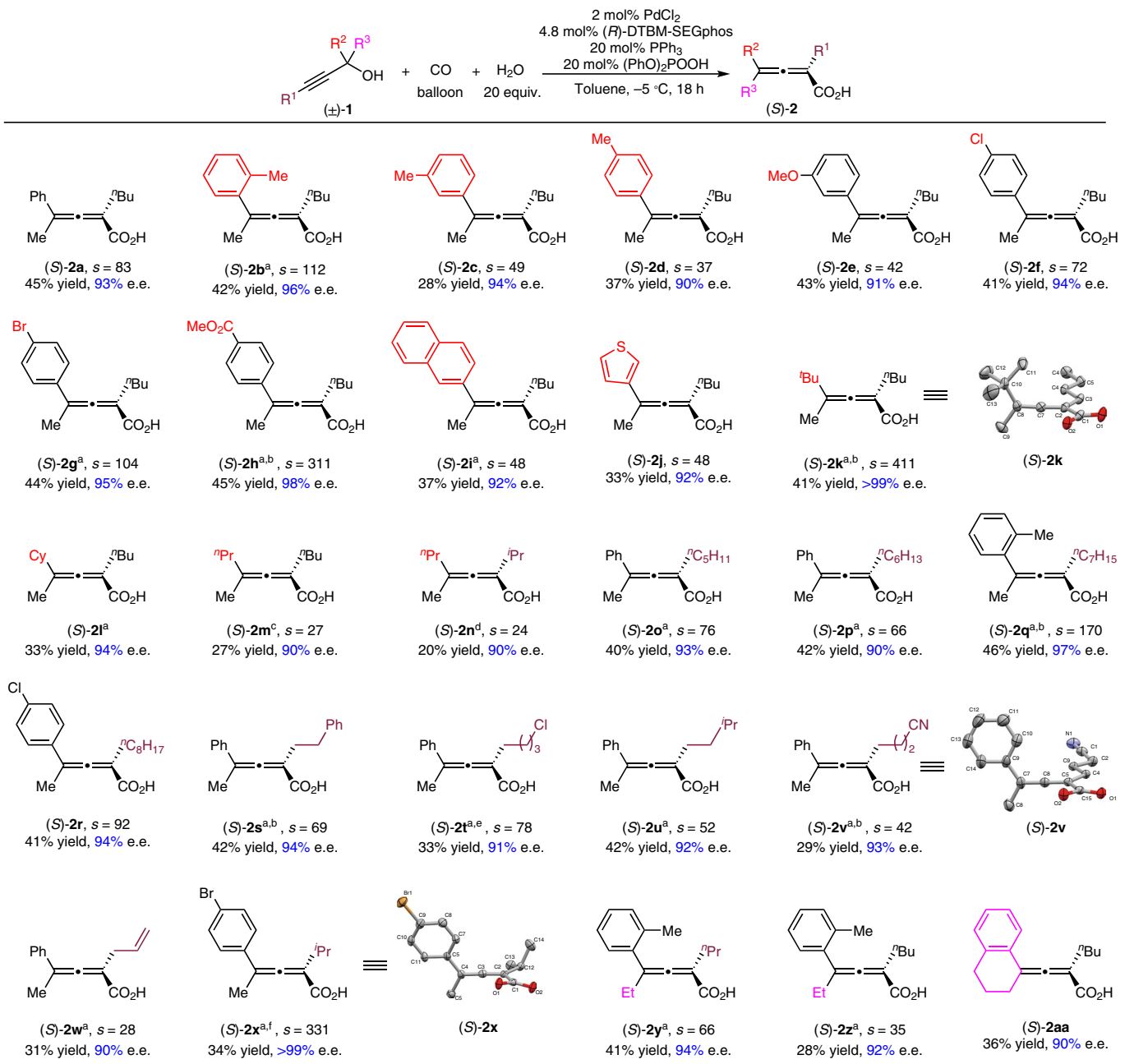
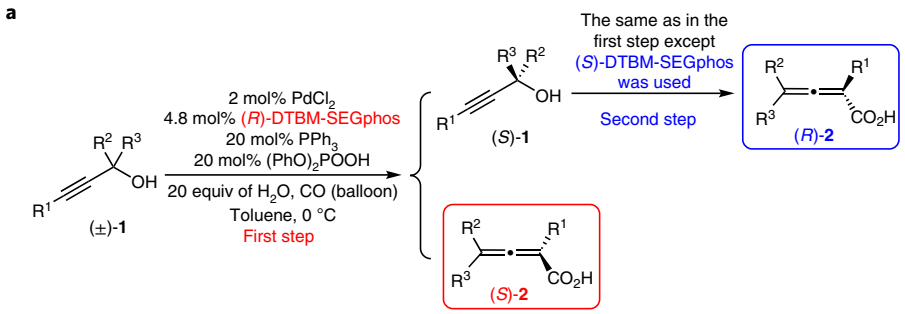


Fig. 2 | Substrate scope study. Reaction conditions: **1** (1.0 mmol), $PdCl_2$ (2 mol%), (*R*)-DTBM-SEGphos (4.8 mol%), PPh_3 (20 mol%), $(PhO)_2POOH$ (20 mol%) and H_2O (20 equiv.) in toluene (5 ml) at −5 °C with a CO balloon unless otherwise noted. *s* is selectivity factor. $s = k_{fast}/k_{slow} = \ln[1-(1-\text{recovery})(1+\text{e.e.})]/\ln[1-(1-\text{recovery})(1-\text{e.e.})]$. Recovery of propargylic alcohol **1** is determined by ^{1}H NMR analysis, and e.e. is the enantiomeric excesses value of the product **2** which is determined by HPLC analysis. [a]The reaction was carried out at 0 °C. [b]The reaction was carried out with 30 mol% $(PhO)_2POOH$. [c]The reaction was carried out at −7 °C. [d]The reaction was carried out at −10 °C. [e]The reaction was carried out with 15 mol% $(PhO)_2POOH$. [f]The reaction was carried out over 30 h.

halide ((*S*)-**2t**), cyano ((*S*)-**2v**), allyl ((*S*)-**2w**), are tolerated. When the methyl group was replaced with ethyl or another cyclic alkyl chain, (*S*)-**2y**, (*S*)-**2z** and (*S*)-**2aa** were obtained in good enantioselectivities. The absolute configuration of the product (*S*)-**2** was established by the X-ray crystallography of (*S*)-**2k**, (*S*)-**2v** and (*S*)-**2x** (Fig. 2).

To further demonstrate the potential of this protocol, we carried out the gram-scale successive kinetic resolution of propargylic alcohols for the synthesis of both enantiomers of chiral 2,3-allenoic acid **2** (Fig. 3a). When 10.0 mmol of (±)-**1a** was submitted to the standard conditions with (*R*)-DTBM-SEGphos as ligand, 0.8529 g (37% yield) of (*S*)-**2a** with 93% e.e. was isolated (Fig. 3a, entry 1). All the recovered (*S*)-**1a** in 63% e.e. was subsequently subjected to the standard conditions, again with the enantiomer (*S*)-DTBM-SEGphos as ligand, to give 0.8172 g (71% yield) of the enantiomer (*R*)-**2a** with 98% e.e. Successive kinetic resolution of other racemic propargylic alcohols, such as (±)-**1y**, (±)-**1f** and all alkyl substituted (±)-**1k**, showed that all were efficiently realized to afford both enantiomers (Fig. 3a, entries 2–4). To demonstrate the practicality of this approach further, a 50 mmol scale successive kinetic resolution of (±)-**1a** was conducted, affording 5.1 g of (*S*)-**2a** and 4.3 g of (*R*)-**2a** (Fig. 3a, entry 5).



Entry	(±)-1	Scale (mmol)	First step: (*S*)-2 Yield, e.e. (%)	First step: (*S*)-1 Yield, e.e. (%)	Second step: (*R*)-2 Yield, e.e. (%)	Total yield of both enantiomers (%)
1	**1a**	10.0	(*S*)-**2a** 37, 93 0.8529 g	(*S*)-**1a** 52, 63 1.0516 g	(*R*)-**2a** 71, 98 0.8172 g	74, 1.6701 g
2	**1y**	10.0	(*S*)-**2y** 39, 93 0.9553 g	(*S*)-**1y** 52, 63 1.1165 g	(*R*)-**2y** 70, 99 0.8785 g	75, 1.8338 g
3	**1f**	10.0	(*S*)-**2f** 41, 95 1.0946 g	(*S*)-**1f** 52, 75 1.2385 g	(*R*)-**2f** 66, 98 0.9115 g	76, 2.0061 g
4	**1k**	10.0	(*S*)-**2k** 46, >99 0.9668 g	(*S*)-**1k** 43, >99 0.7841 g	(*R*)-**2k** 80, >99 0.7248 g	80, 1.6916 g
5	**1a**	50.0	(*S*)-**2a** 45, 93 5.1316 g	(*S*)-**1a** 51, 69 5.1163 g	(*R*)-**2a** 74, 96 4.3055 g	82, 9.4371 g

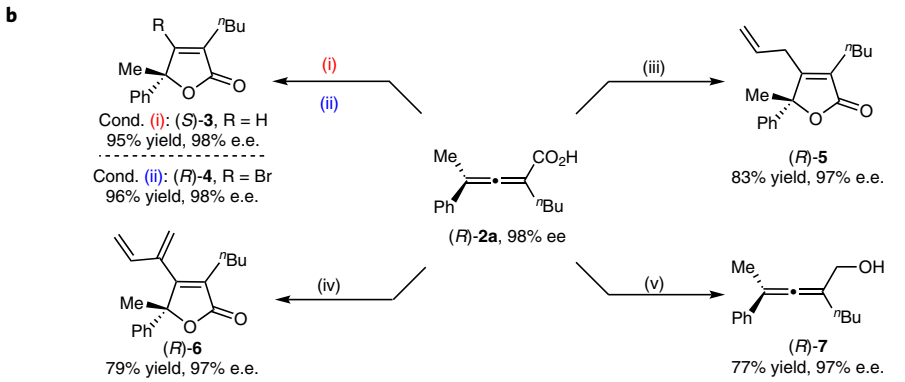


Fig. 3 | Gram-scale reactions and synthetic applications. a, The 10 or 50 mmol scale successive kinetic resolution of racemic propargylic alcohols. **b**, Derivatization of 2,3-allenoic acid (*R*)-**2a**. Reagents and conditions: (i) 4 mol% CuCl, MeOH, 60 °C, 10 h; (ii) 1.2 equiv. *N*-bromosuccinimide (NBS), $CHCl_3$, r.t., 2 h; (iii) 5 mol% $PdCl_2$, 6 equiv. allyl bromide, *N*,*N*-dimethylacetamide (DMA), 50 °C, 20 h; (iv) 5 mol% $PdCl_2$, 2.5 equiv. 2,3-butadien-1-ol, 0.8 equiv. trifluoroacetic acid (TFA), DMA, 30 °C, 8 h; (v) 1.5 equiv. CH_3I, 1.5 equiv. K_2CO_3, DMF, −5 °C, 1.5 h; 2.1 equiv. diisobutylaluminium hydride (DIBAL-H), toluene, −78 °C, 4 h.

Synthetic applications. The synthetic potentials of these chiral 2,3-allenoic acid products have been demonstrated by providing efficient chirality transfer-based protocols to afford optically active compounds with the challenging quaternary chiral carbon centres (Fig. 3b). A cyclization reaction of (*R*)-**2a** catalysed by CuCl proceeded smoothly, affording 2(5*H*)-furanone (*S*)-**3** in 95% yield with 98% e.e.[38]. The bromolactonization of (*R*)-**2a** to 4-bromo-2(5*H*)-furanone (*R*)-**4** (96% yield, 98% e.e.) was successfully realized in the presence of NBS. With $PdCl_2$ as catalyst, the allylated butenolide (*R*)-**5** can be effectively assembled in 83% yield with 97% e.e.[39]. 4-(1′,3′-Dien-2′-yl)-2(5*H*)-furanone derivative (*R*)-**6** was formed with excellent chirality transfer via $PdCl_2$-catalysed cross-coupling cyclization of (*R*)-**2a** with 2,3-allenols[40]. The synthetically versatile tetrasubstituted 2,3-allenol (*R*)-**7** was easily prepared by esterification followed by reduction with DIBAL-H at −78 °C (Fig. 3b).

Mechanistic studies. Kinetic studies revealed a first-order dependence of the reaction rate with respect to substrate **1a** ($v = k \cdot [\mathbf{1a}]$) ([**1a**] is the concentration of substrate **1a** at time *t*) (Fig. 4a; for details see Supplementary Tables 9 and 10). Furthermore, when submitting optically active (*R*)-**1a** and (*S*)-**1a** to the same optimal reaction conditions, separately, (*R*)-**1a** was smoothly converted to the allenoic acid (*S*)-**2a**; however, the reaction of (*S*)-**1a** was extremely sluggish (Fig. 4b; for details see Supplementary Tables 11 and 12), offering an easy-to-handle protocol for the successive kinetic resolution shown in Fig. 3a.

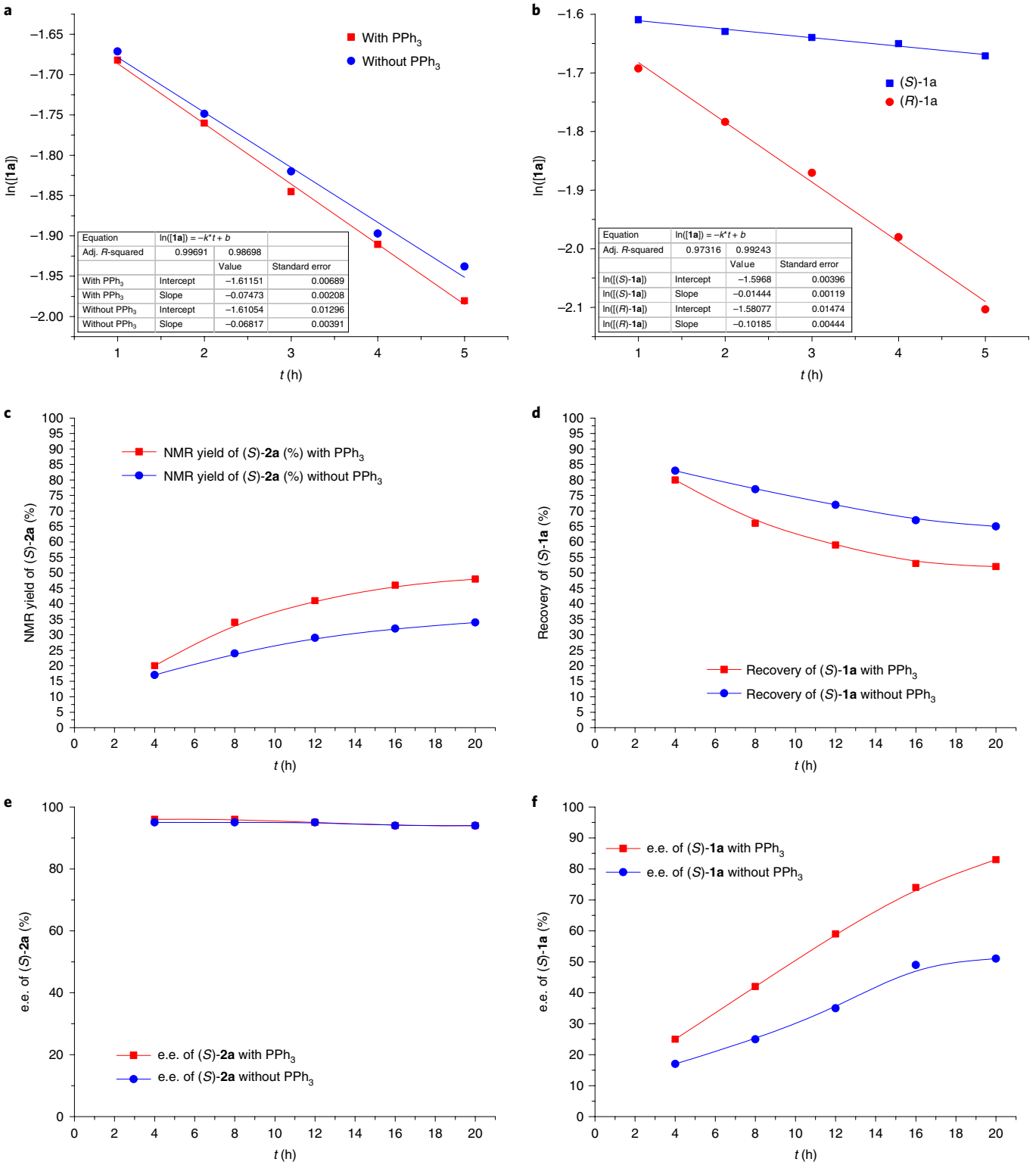
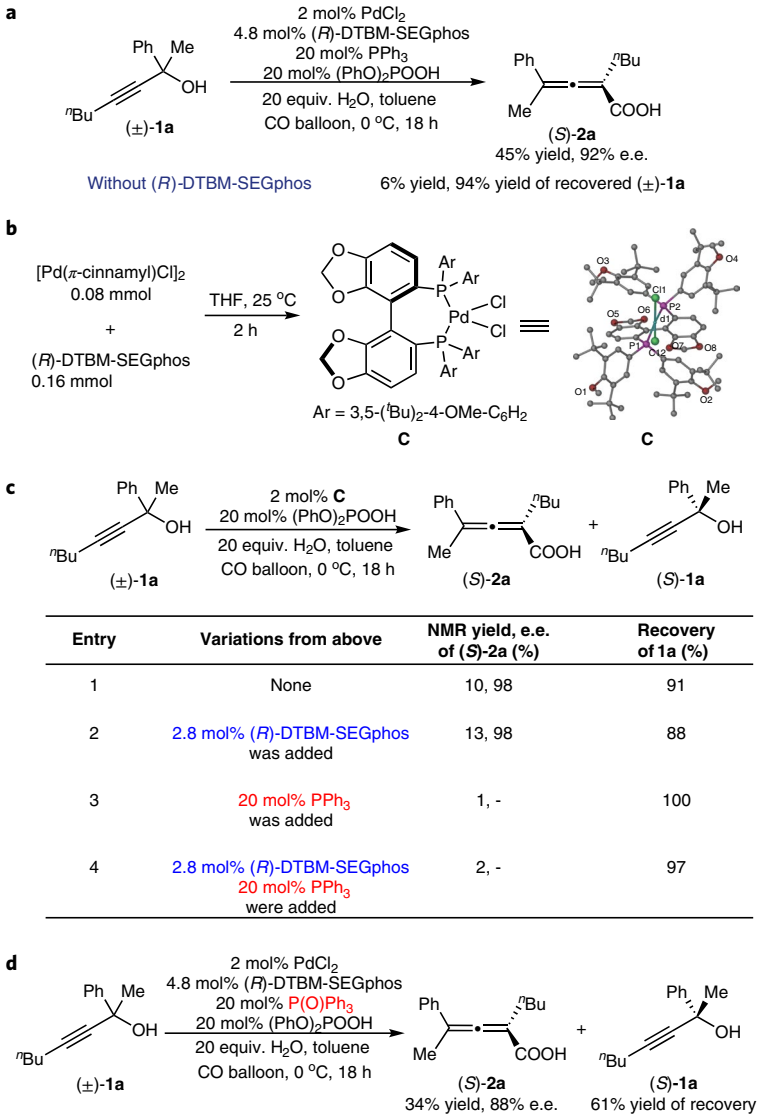


Fig. 4 | Kinetic studies and the reaction profile as monitored by ^{1}H NMR and HPLC. a, The reaction order on substrate **1a**. **b**, The reaction order on substrates (*R*)-**1a** and (*S*)-**1a**. **c**, The effect of PPh_3 on the formation of (*S*)-**2a**. **d**, The effect of PPh_3 on the formation of (*S*)-**1a**. **e**, The effect of PPh_3 on the e.e. of (*S*)-**2a**. **f**, The effect of PPh_3 on the e.e. of (*S*)-**1a**.

To unveil the real role of PPh_3, we monitored the parallel reactions with or without PPh_3 at 0 °C by ^{1}H NMR and HPLC analysis (for details see Supplementary Tables 13 and 14). The rates for the (*S*)-**2a** formation for these two reactions are very similar during the first 4 h. After that, the reaction with the help of PPh_3 was faster (Fig. 4c,d). But the enantioselectivities stayed at the same level during the whole transformation for the two reactions (Fig. 4e), which is different from the reaction conducted at 50 °C shown in entry 4 of Table 1. In addition, the e.e. of the recovered (*S*)-**1a** was higher in the reaction with PPh_3 (Fig. 4f).



Entry	Variations from above	NMR yield, e.e. of (*S*)-2a (%)	Recovery of 1a (%)
1	None	10, 98	91
2	2.8 mol% (*R*)-DTBM-SEGphos was added	13, 98	88
3	20 mol% PPh_3 was added	1, -	100
4	2.8 mol% (*R*)-DTBM-SEGphos 20 mol% PPh_3 were added	2, -	97

Fig. 5 | Control experiments. a, The effect of (*R*)-DTBM-SEGphos on the formation of (*S*)-**2a**. **b**, Preparation of the chiral Pd((*R*)-DTBM-SEGphos)Cl_2 **C**. **c**, The effect of (*R*)-DTBM-SEGphos and PPh_3 on the formation of (*S*)-**2a** with Pd((*R*)-DTBM-SEGphos)Cl_2 **C** as the catalyst. **d**, The effect of P(O)Ph_3 on the formation of (*S*)-**2a**.

To further understand the role of (*R*)-DTBM-SEGphos and PPh_3, a series of control experiments were performed (Fig. 5). The reaction in the absence of (*R*)-DTBM-SEGphos was very slow, affording only 6% yield of (±)-**2a** with 94% yield of recovered (±)-**1a** (Fig. 5a). We also prepared the chiral Pd((*R*)-DTBM-SEGphos)Cl_2 **C** (ref. [41]) by mixing [Pd(π-cinnamyl)Cl]$_2$ with (*R*)-DTBM-SEGphos (Fig. 5b). However, only a 10% NMR yield of (*S*)-**2a** was observed after 18 h when the chiral palladium(II) complex **C** was applied as the pre-catalyst (Fig. 5c, entry 1). There was no significant change when an extra 2.8 mol% of (*R*)-DTBM-SEGphos was added (the total amount of chiral ligand is equal to that in our standard conditions) (Fig. 5c, entry 2). When 20 mol% PPh_3 was used in the reaction, the carboxylation reaction was suppressed (Fig. 5c, entries 3 and 4). These results suggest that once the chiral bidentate ligand (*R*)-DTBM-SEGphos was fully coordinated to $PdCl_2$, the addition of supporting ligand PPh_3 would have a negative effect on the reaction. When P(O)Ph_3 was applied, the (*S*)-**2a** was formed in 34% yield with 88% e.e. (Fig. 5d). This result excludes the possibility that PPh_3 was in situ converted to phosphine oxide to affect the carboxylation reaction.

$PdCl_2$, (*R*)-DTBM-SEGphos and PPh_3 were then mixed in DCM and C_6D_6. In the ^{31}P NMR spectrum of the resulting mixture, the signals of (*R*)-DTBM-SEGphos ($\delta = -13.7$ ppm), PPh_3 ($\delta = -5.7$ ppm), $Pd(PPh_3)_2Cl_2$ ($\delta = 23.4$ ppm), palladium complex **C** Pd((*R*)-DTBM-SEGphos)Cl_2 ($\delta = 28.9$ ppm) and four new signals ($\delta = 32.4$, 27.1, 23.9, 19.3 ppm) (for details, see Supplementary Fig. 2) were observed. The signals of phosphine oxides were not observed.

This same resulting mixture was then used for solvent-assisted electrospray ionization mass spectrometry (SAESI-MS) studies[42]. A signal with *m*/*z* of 1,619.1, which matched the *m*/*z* of [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 + H]$^+$ (Fig. 6a, calcd for $C_{92}H_{116}{}^{35}Cl_2O_8P_3{}^{106}Pd^+$: 1,619.6), was observed. For further structural characterization of this intermediate, a SAESI-MS/MS experiment was conducted: the corresponding signals of [Pd((*R*)-DTBM-SEGphos)Cl_2]$^+$ (Fig. 6b, calcd for $C_{74}H_{100}{}^{35}Cl_2O_8P_2{}^{106}Pd^+$: 1,356.5), formed by losing H$^\bullet$ and one molecule of PPh_3, [Pd((*R*)-DTBM-SEGphos)Cl]$^+$ (Fig. 6b, calcd for $C_{74}H_{100}{}^{35}ClO_8P_2{}^{106}Pd^+$: 1,321.6), formed by losing one molecule each of HCl and PPh_3, and [(*R*)-DTBM-SEGphos + H]$^+$ (Fig. 6b, calcd for $C_{74}H_{101}O_8P_2^+$: 1,179.7) were detected. In addition, the signal with *m*/*z* of 1,583.8,

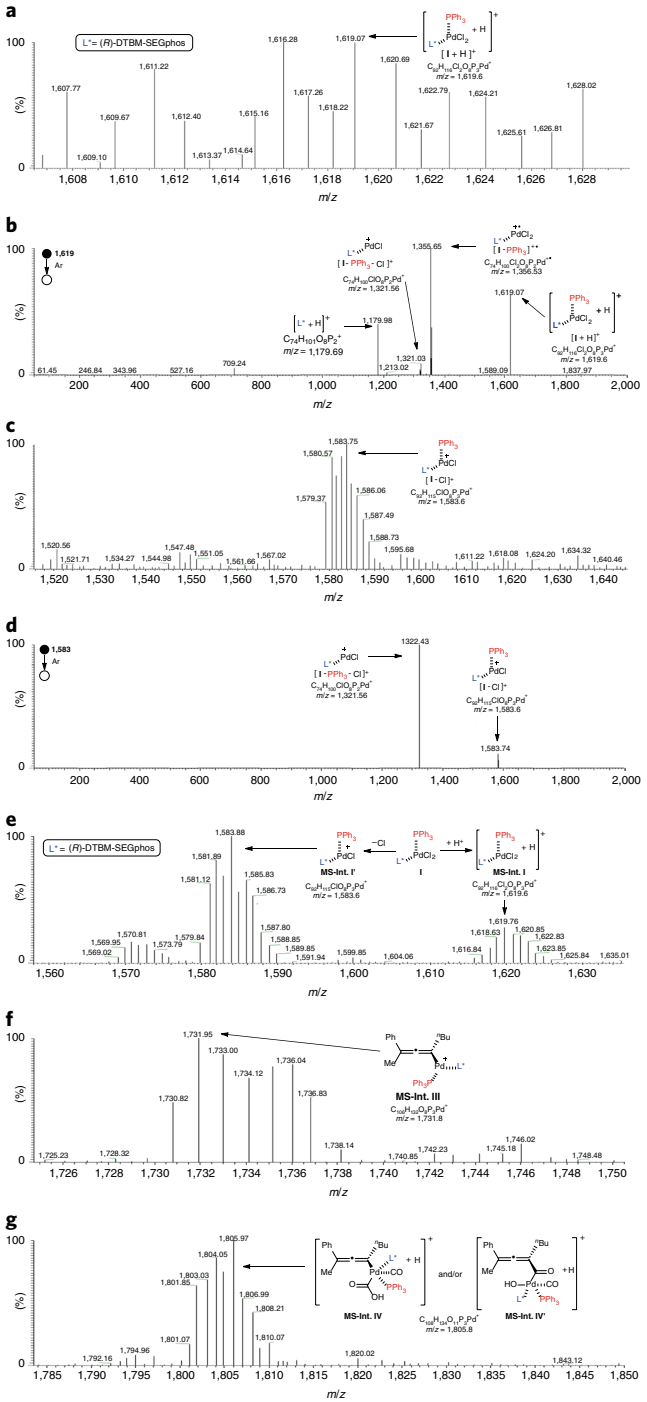


Fig. 6 | SAESI-MS studies. a, SAESI-MS spectrum of the solution showing the signal of [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 + H]$^+$ [**I** + H]$^+$ at *m/z* 1,619. **b**, SAESI-MS/MS spectrum of complex ion [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 + H]$^+$ [**I** + H]$^+$ at *m/z* 1,619. **c**, SAESI-MS spectrum of the solution showing the signal of [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl]$^+$ [**I** − Cl]$^+$ at *m/z* 1,583. **d**, SAESI-MS/MS spectrum of complex ion [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl]$^+$ [**I** − Cl]$^+$ at *m/z* 1,583. **e**, Expanded SAESI-MS spectrum, showing the signal from *m/z* 1,560 to 1,650. **f**, Expanded SAESI-MS spectrum, showing the signal from *m/z* 1,720 to 1,750. **g**, Expanded SAESI-MS spectrum, showing the signal from *m/z* 1,785 to 1,850.

which belongs to [Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl]$^+$, formed from Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 by losing a chloride anion, was also detected (Fig. 6c, calcd for $C_{92}H_{115}{}^{35}ClO_8P_3{}^{106}Pd^+$: 1,583.6). Its identity was further confirmed by the SAESI-MS/MS experiment, during which one molecule of PPh_3 was dissociated (Fig. 6d, calcd for $C_{74}H_{100}{}^{35}ClO_8P_2{}^{106}Pd^+$: 1,321.6). Based on the data shown in Supplementary Fig. 2 and Fig. 6a–d and the coordination ability of Pd, we concluded the existence of precatalyst Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 **I**, in which (*R*)-DTBM-SEGphos was mono-ligated; thus, in the ^{31}P NMR spectra (Supplementary Fig. 2), the doublets at 30.6 and 25.5 ppm with J_{PP} = 545.5 Hz may be assigned to the ligated P atom in (*R*)-DTBM-SEGphos and the ligated PPh_3 in this species.

To further understand the role of the supporting ligand PPh_3 in the reaction process, we also carried out SAESI-MS studies of the real reaction mixture with everything inside (Fig. 6e–g, for details see Supplementary Figs. 15–20). A solution of 2-phenyloct-3-yn-2-ol **1a** (1 mmol), $PdCl_2$ (0.02 mmol), (*R*)-DTBM-SEGphos (0.048 mmol), PPh_3 (0.2 mmol), $(PhO)_2POOH$ (0.2 mmol) and H_2O (20 mmol) in toluene (5 ml) was stirred at −5 °C in a CO atmosphere provided with a balloon. After 10 min, the pre-catalyst Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 **I** was detected again (Fig. 6e). In addition to this, the palladium intermediate **MS-Int. III** (formed via oxidative addition of protonated **1a** with the in situ generated Pd(0)((*R*)-DTBM-SEGphos)(PPh_3) **II** via the reduction of **I** (Fig. 6f)), the CO-inserted non-classic penta-coordinated intermediate **MS-Int. IV** and/or **MS-Int. IV′** bearing both (*R*)-DTBM-SEGphos and PPh_3 were detected (Fig. 6g).

Based on these studies and previous control experiments, a mechanism is proposed as shown in Fig. 7a. Pd((*R*)-DTBM-SEGphos)(PPh_3)Cl_2 **I** is the pre-catalyst, which would be reduced in situ to form the catalytically active Pd(0)((*R*)-DTBM-SEGphos)(PPh_3) **II**. Its S_N2'-type anti-oxidative addition with a Brønsted acid activated intermediate, protonated propargylic alcohol (*R*)-**1′**, would yield the allenylpalladium intermediate **MS-Int. III**, which would react with CO and water to afford the carbonylation intermediate **IV** or **IV′**. Subsequent reductive elimination would afford the (*S*)-allenoic acid and regenerate the catalyst. An alternative all Pd(II)-catalytic cycle is depicted in Fig. 7b: the Pd(II)L*(PPh_3) species would coordinate with the protonated propargylic alcohol forming intermediate **V**, which would further form intermediate **VI** in the presence of CO and H_2O. Its insertion reaction would yield intermediate **VII**. The desired 2,3-allenoic acid would be formed by the β-O elimination of intermediate **VII**. However, no such intermediates **V**–**VII** were detected by SAESI-MS studies. In addition, with the preformed $Pd_2(dba)_3{\cdot}CHCl_3$ as the catalyst, 2,3-allenoic acid (*S*)-**2a** was also obtained in 42% yield with 90% e.e. (Supplementary equation (2)). In conclusion, the Pd(0)–Pd(II)–Pd(0) is more possible (Fig. 7a).

Conclusions

In summary, we have realized the efficient catalytic asymmetric formation of tetrasubstituted 2,3-allenoic acids from very common racemic propargylic alcohols. Enabled by the co-catalysis of palladium and a Brønsted acid, the successive kinetic resolution reaction of propargylic alcohols proceeded efficiently under 1 atm CO and water with a very decent functional group compatibility, affording both enantiomers of these tetrasubstituted 2,3-allenoic acids in excellent enantioselectivity and atom economy. The presence of PPh_3 as a supporting ligand is critical for this protocol and such a strategy may also work in other metal-catalysed asymmetric catalysis. Such chiral 2,3-allenoic acids are very useful in organic chemistry. Further studies in this area, including investigation of the working mechanism for PPh_3, are being actively pursued in this laboratory.

Methods

General procedure for the palladium-catalysed enantioselective carboxylation reaction. To a Schlenk flask (25 ml) were added $PdCl_2$ (3.6 mg, 0.02 mmol), (*R*)-DTBM-SEGphos (57.0 mg, 0.048 mmol), PPh_3 (52.7 mg, 0.2 mmol) and $(PhO)_2PO_2H$ (50.1 mg, 0.2 mmol). After the addition of all solid chemicals, the flask

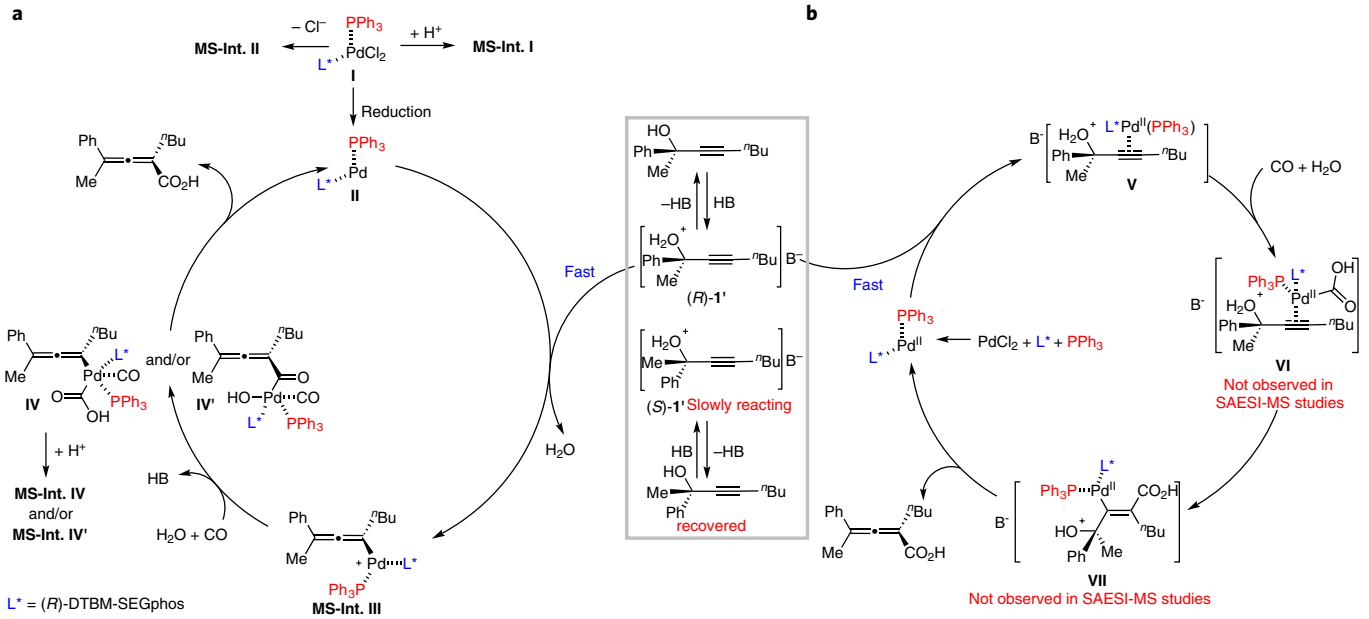


Fig. 7 | Proposed mechanisms. a, The more possible mechanism. **b**, An alternative mechanism.

was degassed and refilled with Ar three times to ensure the complete exclusion of air. Then (±)-**1a** (201.6 mg, 1.0 mmol)/toluene (3 ml), H_2O (360 µl, $d = 1.0\ g\ ml^{-1}$, 360.0 mg, 20 mmol) and toluene (2 ml) were added sequentially. The Ar gas line was then closed. The resulting mixture was frozen with a liquid nitrogen bath, degassed to completely remove the argon inside, and refilled with CO by a balloon of CO (~1 l) three times. The liquid nitrogen bath was removed and the resulting mixture was allowed to stand until completely thawed, vigorously stirred at −5 °C with a balloon of CO for 18 h, warmed up to room temperature, treated with H_2O_2 (40 µl, $d = 1.13\ g\ ml^{-1}$, 30 wt% in H_2O, 13.5 mg, 0.4 mmol), stirred for 30 min at room temperature, diluted with 5 ml of ethyl acetate, filtered through a short column of silica gel (3 cm) eluted with ethyl acetate (20 ml) and concentrated. The residue was purified by chromatography on silica gel to afford the pure product.

Data availability

The X-ray crystallographic coordinates for the structures of (*S*)-**2k**, (*S*)-**2v**, (*S*)-**2x** and Pd((*R*)-DTBM-SEGphos)Cl_2 **C** reported in this Article have been deposited at the Cambridge Crystallographic Data Centre (CCDC) under deposition numbers 1891253 ((*S*)-**2k**), 1855080 ((*S*)-**2v**), 1854910 ((*S*)-**2x**) and 1855156 (Pd((*R*)-DTBM-SEGphos)Cl_2 **C**). These data can be obtained free of charge from http://www.ccdc.cam.ac.uk/data_request/cif. Experimental procedures and characterization of the new compounds are available in the Supplementary Information. All other data are available from the authors upon reasonable request.

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Author contributions

S.M. and H.Q. directed the research and developed the concept of the reaction with W.-F.Z., who also performed the experiments and prepared the Supplementary Information. W.Z., C.H. and P.W. were involved in the synthesis of substrates. Y.-L.G. directed the SAESI-MS reaction with L.W., who also collected and analysed the SAESI-MS spectra data. W.-F.Z., H.Q. and S.M. checked the experimental data. W.-F.Z., H.Q. and S.M. wrote the manuscript, with contributions from the other authors.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to H.Q., Y.-L.G. or S.M.

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